Self-Supervised Contrastive Learning for Epileptic Seizure Prediction

Integrating Explainable AI for EEG Signal Interpretation

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Motivation: Bridging the Gap in Seizure Prediction

- **Data Inefficiency in Clinical EEG:**
  - Expert annotation is expensive, slow, and labor-intensive.
  - Pre-ictal states are subtle and vary significantly across patients.
- **The "Black Box" Dilemma:**
  - High accuracy alone does not guarantee clinical adoption.
  - Neurologists require physiological rationale to trust AI decisions.

Why This Matters

Integrating **Self-Supervised Contrastive Learning** with **Explainable AI** enables robust prediction from unlabeled data while ensuring transparency for medical practitioners.

Background: EEG Dynamics and Seizure States

- **Electroencephalogram (EEG) Characteristics:**
  - Non-invasive recording of cortical electrical activity.
  - High temporal resolution but extremely low signal-to-noise ratio.
- **The Transition Timeline:**
  - *Inter-ictal:* Baseline normal brain activity.
  - **Pre-ictal:** The critical transition period before a seizure.
  - *Ictal:* The active seizure event.

The Scientific Challenge

Identifying the **Pre-ictal** state is a "needle in a haystack" problem. The signal changes are non-linear, non-stationary, and often imperceptible to traditional visual inspection.

Problem Statement and Research Goal

The Problem:

- **Label Scarcity:** Current models rely on massive, expert-annotated datasets, making clinical scaling difficult and expensive.
- **Complex Dependencies:** Traditional architectures often neglect the simultaneous spatio-temporal-spectral correlations inherent in EEG signals.
- **Opaque Decision Making:** Most high-performing models function as "black boxes," hindering medical trust and clinical adoption.

Research Goal

To develop a **Self-Supervised Contrastive Learning** framework that extracts robust features from unlabeled EEG data, integrated with **Explainable AI** to provide neurologists with interpretable physiological evidence for seizure predictions.

Proposed Method I: Self-Supervised Representation

- **Data Augmentation & Tuple Construction:**
  - Raw EEG is transformed into time-frequency patches using **STFT** or **Wavelet Transforms**.
  - Pre-text tasks: Temporal shuffling, channel masking (80%), and adding Gaussian noise to create *Positive* and *Negative* pairs.
- **Contrastive Learning Objective:**
  - A **Dual-Pyramid Spatio-Temporal-Spectral** backbone encodes these pairs into a latent space.
  - **InfoNCE Loss** is utilized to maximize mutual information between augmented versions of the same EEG segment.

The Technical Advantage

By training on unlabeled data first, the model learns the intrinsic spatio-temporal structure of EEG without needing expensive neurologist-labeled datasets.

Proposed Method II: Prediction & Explainability

- **Adaptive Attention Backbone:**
  - Features are passed through a **Spatio-Temporal-Spectral Transformer**.
  - **Triple Attention Layers (TAL)** weigh the importance of specific channels (Spatial) and frequency rhythms (Spectral).
- **Interpretability via XAI:**
  - **Grad-CAM** visualization to highlight the most influential EEG sensors.
  - **Attention Maps** provide a "heatmap" of the pre-ictal window, showing which spikes triggered the alarm.

Clinical Utility

The model outputs both a **Seizure Probability Score** and a **Visual Evidence Map**, allowing clinicians to verify predictions against physiological biomarkers.

Dataset Specifications

- **CHB-MIT Scalp EEG (Boston Children's Hospital):**
  - **Population:** 22 pediatric subjects (5 males, ages 3–22; 17 females, ages 1.5–19).
  - **Volume:** 969 hours of recordings across 664 EDF files.
  - **Seizure Events:** 198 total annotated seizures.
  - **Hardware:** 256 Hz sampling rate, 16-bit resolution, 23–26 channels (10-20 system).
- **TUH EEG Corpus (Temple University Hospital):**
  - **Total Size:** $>$30,000 clinical sessions from 14,000+ patients.
  - **Seizure Subset (TUSZ v2.0.3):** 7,377 files from 675 subjects (1,476 total hours).
  - **Pre-training Utility:** High volume of unlabeled data for **Contrastive Learning**.
  - **Hardware:** $\geq$250 Hz sampling rate, minimum 17 EEG channels.

Key Contributions and Impact

- **Main Technical Improvements:**
  - **Learning from Raw Data:** The model uses large amounts of unlabeled EEG to improve accuracy without needing extra manual work from doctors.
  - **Smart Signal Analysis:** Uses **Triple Attention** and **Pyramid Networks** to look at brain waves, timing, and location all at once.
  - **Brain Mapping:** Uses **Dynamic Graphs** to track how different parts of the brain communicate before a seizure.
- **How This Helps Patients and Doctors:**
  - **Interpretability:** Provides visual and clinical evidence to support model predictions, making the system more reliable for medical use.
  - **Efficiency:** Optimized design suitable for real-time monitoring on mobile or wearable healthcare devices.

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Thank You!

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